

In addition to the required standard motor starter fitting-out mentioned above, frequency converter motor starters shall be performed according to the following.

All frequency inverters shall be selected heavy-duty type. All panels shall be designed to isolation contactor before than frequency inverters. All frequency inverter communicate PLC by the Ethernet protocols.

Earth leakage protection shall be incorporated either using electronic relays or by adding residual current protection to the motor circuit breaker. The earth leakage protection shall in either cases trip the motor circuit breaker instantaneously in the event of an insulation fault in the motor circuitry, ensuring that the system-to-frame voltage will not exceed 50 V AC and will be cleared within 5 sec, as required pursuant to IEC 60364. Attention is drawn to the fact that the earth leakage protection must be of a type, which is capable of functioning with frequency converters.

It shall be heavy duty type. fk series shall be provided if it'll be used for pumps.

If the frequency converters employed are incorporated with overload protection the latter may be utilised for motor overload protection in lieu of thermistor protection relays. It will be used in the event that motor system maintains this requirement. (except thermal protection and water leakage signals which provided from manufacturer).

Each frequency converter shall be clearly labelled with the number of the motor/pump, which it is serving. The frequency converter shall be placed so its internal control panel is placed approximately 1600 mm above finished floor level. If installed inside an assembly provision shall be made for adequate ventilation.

When the panel hand-off-automatic (HOA) selector switch is in position (signal also to PLC):

- Hand: The operator controls operation of the motor from the front of the frequency converter. The automatic control system is bypassed.
- Off: The motor is switched off and inoperative. The automatic control system is bypassed.
- Automatic (Auto): The motor is controlled by the automatic control system.

The start/stop push buttons, the required specification for a standard motor controller shall not to be employed. Manual control shall be performed from the front of each frequency converter where the operator shall be capable of starting/stopping and turning up and down the pump revolutions.

Each frequency converter shall be configured to display the following current values on request:

- Ampere [A]
- Power factor (0-1)
- True power [kW]
- Apparent power [kVA]
- Revolutions per minute [RPM]

Frequency converters shall comply with:

- EN 60439, Low-voltage switchgear and control gear assemblies.
- IEC 60664, Insulation co-ordination for equipment within low-voltage systems.
- EN 50082-1, Electromagnetic compatibility (EMC) genetic immunity standard part 1: Industrial environment.

Frequency converters shall be 3 phase - 400 V grade, equipped with EMC-filter, fully enclosed to a degree Frequency converters to be used inside panel will have minimum IP41 and ones to be used directly in outdoors